

Notes.

(a) You may freely use any result proved in class unless you have been asked to prove the same. Use your judgement. All other steps must be justified.

(b) We use $\mathbb{R}$ = real numbers, $I = [0, 1]$, the unit interval in $\mathbb{R}$, S^n = the unit sphere in $\mathbb{R}^{n+1}$.

(c) There are a total of **110** points in this paper. You will be awarded a maximum of **100** points.

1. [8 + 12 = 20 Points] Let X be a topological space, and let $x_0, x_1 \in X$.

(i) For any path $\alpha: I \rightarrow X$ with $\alpha(0) = x_0$ and $\alpha(1) = x_1$, describe the induced homomorphism $\hat{\alpha}: \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ (give only the description and not the well-definedness of $\hat{\alpha}$ or why it is a homomorphism) and explain why it is bijective.

(ii) Give an example where two different choices of the path, say α, β , lead to two different induced maps $\hat{\alpha}, \hat{\beta}$. (You may use any result or example covered in class.)

2. [15 Points] Let M be a connected topological manifold of dimension $n > 2$. Let $x_0 \in M$. Prove that for any other point $p \in M$, we have $\pi_1(M, x_0) \cong \pi_1(M \setminus \{p\}, x_0)$.

3. [30 Points] Construct a simplicial complex K such that its polytope $|K|$ is homeomorphic to the compact cylinder $S^1 \times I$. Compute its homology groups.

4. [20 Points] Let $f, g: K \rightarrow L$ be simplicial maps satisfying the following condition: *For any simplex $\sigma = v_0, \dots, v_p$ in K , the simplex on the vertices $\{f(v_0), \dots, f(v_p)\}$ and that on the vertices $\{g(v_0), \dots, g(v_p)\}$ both lie in an acyclic subcomplex $L(\sigma)$ of L .* Moreover, $\tau \subset \sigma \implies L(\tau) \subset L(\sigma)$.

Prove that there is a chain homotopy between $f_\#$ and $g_\#$.

5. [5 + 5 + 5 + 10 = 25 Points] Let $h: |K| \rightarrow |L|$ be a continuous map between polytopes of simplicial complexes K, L in $\mathbb{R}^N$.

(i) Define what it means for h to satisfy the star condition.

(ii) Define what is meant by a simplicial approximation to h .

(iii) If K and L are finite, prove that $|K|$ and $|L|$ have the subspace topology of $\mathbb{R}^N$.

(iv) Under the assumptions in (iii), assume further that h admits a simplicial approximation $f: K \rightarrow L$. Show that there exists a continuous homotopy between f and h , both viewed as maps from $|K| \rightarrow |L|$. (You may assume the basic facts on simplicial approximation proved in class).